



CMPT 305 Project Presentation

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Target Users

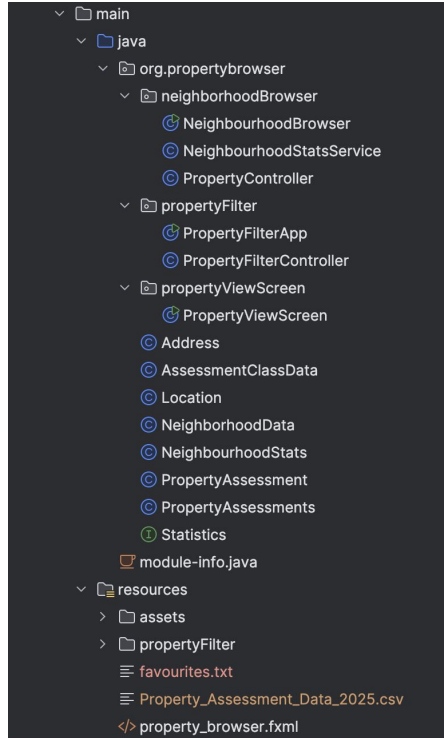
- Target users for our application are
 - Home buyers (Primarily)
 - Investors
 - Surveyors
 - New residents coming from different cities
- Identified by the questions of “Who can we serve with the data we have?” and “What kind of tool would people want to use?”

Target Users

Why use our app?

- Moving to Edmonton, looking for a neighbourhood leveraging safety and affordability?
 - Or any other possible factors: Average costs within neighbourhood, population density, local amenities?
- Trying to move to a new neighbourhood, looking for specific things in a future home?
- Want an idea of what it looks like in an area?
- Easy comparison between distinct properties
- Easy lookup of properties you're familiar with
- User-friendly saving and revisiting properties for comparison and analysis

Project Implementation



- Three screens:
 - Property browser
 - Property Viewer
 - Neighborhood Information Viewer
- Organized into packaged for each screen, and resources folder containing FXMLs for UI implementation
- Uses the Edmonton Property Assessment Open Data for September 2025
- Accessed within the resources folder

Design Decisions

- Biggest questions we faced:
 - What does the user want/don't want?
 - What data do we show and leave out?
 - How do we implement what we want?
 - Layout was the first thing planned, each element had its own questions
- For the map, we debated between
 - ArcGIS embedded map
 - Gluon embedded map
 - Google Maps embedded map
- Ultimately went with Google Maps to distinguish ourselves
 - Native javascript embed was attempted, but was difficult to implement
 - WebView offered a much easier way to bring it in, but has its own quirks

Design Questions

- What kind of functions would a user want, and what would they not want?
- How can we provide users with an easy to use, intuitive, and comprehensive way to search property listings?
- Can we add multiple ways for a user to complete a task?
 - For example: searching for properties by pressing a button OR using a shortcut key
- Can we make the interface dynamic based on user actions?
 - For example:
 - hiding the neighborhood box when not required
 - switching between tabs on the table automatically

Browser Screen: Original Mockup

Property Viewer App

Property address text query

Dynamic dropdown (addresses)

- Original design included only an option to browse by matching address lines
- The addresses would appear in a dropdown ComboBox and update dynamically as the user types

Browser Screen: Original Mockup (Issues)

Property Viewer App

Property address text query

Dynamic dropdown (addresses)

Issues with this approach:

- **Too restrictive:** lacked functionality in allowing users to browse properties in a variety of different ways
- **Too slow:** At low levels of filtering (i.e. when the user begins by typing 1 or 2 letters), the number of addresses retrieved in the dropdown is very high, leading to slowdowns and program performance degradation

Browser Screen: Alternative Solutions

Property Viewer App

Property address text query

Dynamic dropdown (addresses)

Possible alternative implementations:

- Limit the number of properties retrieved in the combobox (i.e. limit = 5)
- Combine the browser (which has minimal functionality) with the property viewer and map into one screen
- A list or combobox to browse properties in a user's history or favorites list

Browser Screen: Final Implementation

Results	History	Favourites		
Address		Value	Neighborhood	Has Garage
2744 1 Avenue Sw		\$184,000	Alces	No
2748 1 Avenue Sw		\$578,000	Alces	Yes
2752 1 Avenue Sw		\$183,500	Alces	No
2756 1 Avenue Sw		\$319,500	Alces	Yes
2804 1 Avenue Sw		\$648,500	Alces	Yes
2808 1 Avenue Sw		\$184,000	Alces	No
2812 1 Avenue Sw		\$184,000	Alces	No
2816 1 Avenue Sw		\$173,000	Alces	No
2820 1 Avenue Sw		\$653,500	Alces	Yes
2824 1 Avenue Sw		\$550,000	Alces	Yes
2828 1 Avenue Sw		\$684,000	Alces	Yes
2832 1 Avenue Sw		\$184,000	Alces	No
2836 1 Avenue Sw		\$173,000	Alces	No
2904 1 Avenue Sw		\$651,500	Alces	Yes
2908 1 Avenue Sw		\$473,000	Alces	Yes
2912 1 Avenue Sw		\$643,000	Alces	Yes
2916 1 Avenue Sw		\$759,000	Alces	Yes
2920 1 Avenue Sw		\$605,500	Alces	Yes
2924 1 Avenue Sw		\$642,000	Alces	Yes
2928 1 Avenue Sw		\$733,000	Alces	Yes
2932 1 Avenue Sw		\$724,000	Alces	Yes
2936 1 Avenue Sw		\$440,000	Alces	Yes
2940 1 Avenue Sw		\$808,000	Alces	Yes

Organization: Displays results on an interactive table for simplicity and clarity

- In TableView, users can sort by any of the column fields (i.e. address, value, neighborhood)

Browser Screen: Final Implementation

Expanded functionality:

- Allows the user to filter properties using a variety of different parameters, or optionally ignore them
- History and Favorites tab reuses existing results table for UI consistency, reducing clutter, and to minimize the need to design new resources
- Added reset search fields button, ENTER to search, and automatic switching to the results tab when searching

Property Browser

Results: 20 properties found

Min Value:

Max Value:

Has Garage: ☒

Filter by Neighborhood: ☒

Neighborhood:

Address:

Reset Search:

Address	Value	Neighborhood	Has Garage
12105 39 Avenue Nw	\$1,472,000	Westbrook Estates	Yes
12107 39 Avenue Nw	\$820,000	Westbrook Estates	Yes
12111 39 Avenue Nw	\$965,500	Westbrook Estates	Yes
12203 39 Avenue Nw	\$880,500	Westbrook Estates	Yes
12207 39 Avenue Nw	\$1,351,000	Westbrook Estates	Yes
12303 39 Avenue Nw	\$1,004,500	Westbrook Estates	Yes
12307 39 Avenue Nw	\$917,500	Westbrook Estates	Yes
12311 39 Avenue Nw	\$901,000	Westbrook Estates	Yes
12407 39 Avenue Nw	\$898,000	Westbrook Estates	Yes
12411 39 Avenue Nw	\$942,000	Westbrook Estates	Yes
12415 39 Avenue Nw	\$1,187,000	Westbrook Estates	Yes
12503 39 Avenue Nw	\$863,000	Westbrook Estates	Yes
12507 39 Avenue Nw	\$818,500	Westbrook Estates	Yes
12511 39 Avenue Nw	\$911,500	Westbrook Estates	Yes
12515 39 Avenue Nw	\$1,127,000	Westbrook Estates	Yes
12603 39 Avenue Nw	\$840,000	Westbrook Estates	Yes
12703 39 Avenue Nw	\$853,000	Westbrook Estates	Yes
12711 39 Avenue Nw	\$829,000	Westbrook Estates	Yes
12715 39 Avenue Nw	\$1,286,500	Westbrook Estates	Yes
12716 39 Avenue Nw	\$1,283,500	Westbrook Estates	Yes

Property View Screen: Original Mockup



Property Viewer

street view graphic

Address

Assessment value

Location coordinates

Garage (Y/N)

Ward

[Neighborhood](#)

google maps

More or less the same as the final product

Extra features were added on to a screen produced in the outline's image

Property View Screen: Alternative Ideas



Property Viewer

street view graphic

Address

Assessment value

Location coordinates

Garage (Y/N)

Ward

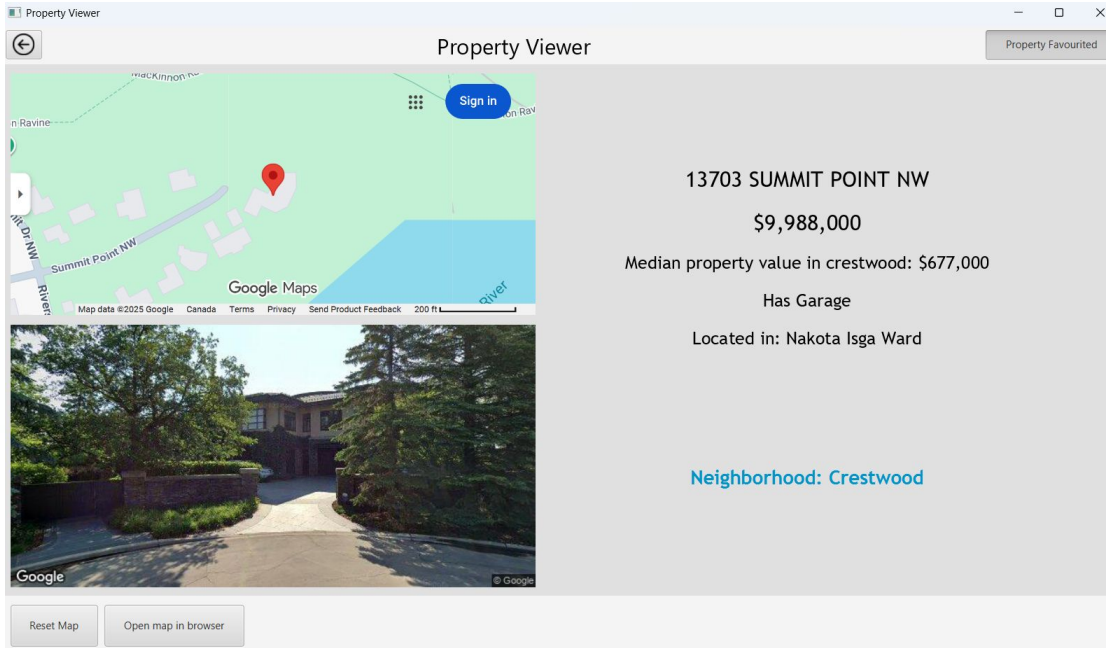
[Neighborhood](#)

google maps

The idea to place multiple properties on one screen was explored

- Created a very busy looking app
- Slightly more technically challenging
- Eventually turned into the “favourites”

Property View Screen: Final Implementation



Product is consistent with original draft

Layout is sensible and has not been difficult to users so far

Visual aids remain on the left while information is written on the right

Property View Screen: Final Implementation



Google Maps is interactive and responsive via WebView

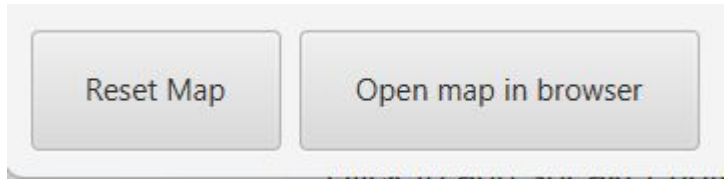
- Places API used to gather a Place ID from the address, which is then loaded on the map
- Street View Static API retrieves an image from latitude and longitude
 - ~20% Of the time the image is of poor quality (blocked camera, blurred location) or no image exists

Property View Screen: Final Implementation

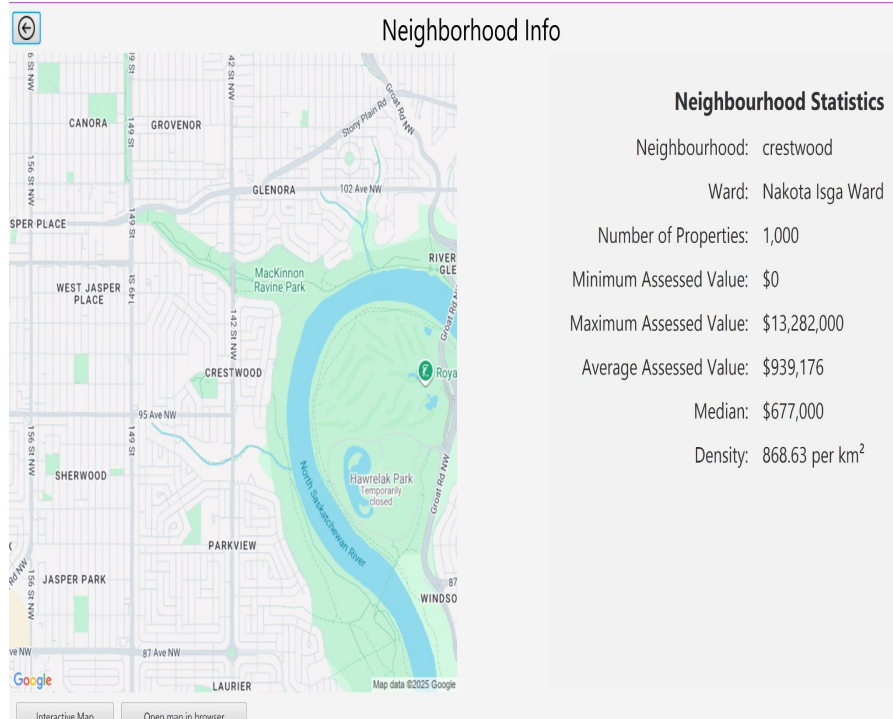
Functional buttons were implemented for useful user tools



- Favourite Property button is a toggle on/off that adds the property to a list of favourites
- Back button returns the user to the search
- Reset Map reloads the map for the user
- Open map in browser allows the user to open the map in their own browser for a larger screen

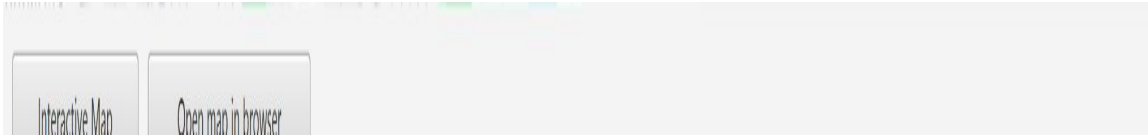


Neighbourhood Screen



Once user clicks on the url from the property view screen (previous screen) they'll be directed to the full on map of the neighborhood and as well as their statistics so homebuyers know the general details of the neighbourhood.

Map Navigation



When user clicks the interactive map button it'll direct them to an overview of the neighborhood off of google. Open browser button will direct you to the overview of both the neighbourhood and map on google maps

Back Button Functionality



Neighborhood Info

If home buyers change their mind about a particular neighbourhood or house they're able to go back to the previous screen that way they're not stuck on the one screen.

Issues and Further Extension - Property Browser

- The editable ComboBox fails to clear its text on the first attempt when the user highlights the input and presses backspace/delete, requiring a second attempt for the text to clear.
- Could work towards implementing a dropdown menu for property addresses as an additional interface option

Issues and Further Extension - Maps Services

- Some properties do not have a retrievable image for Street View
 - Or the image is blurry, blocked by an object, or otherwise not helpful
- Webview has graphical incompatibilities with google maps
 - On larger display sizes, certain elements are loaded that cause graphical issues that slow the application
 - Inconsistently, unsure what causes this
 - Street View in the static Google Maps is not interactive
 - May be unintuitive considering the previous screen began with an interactive map